

Can word ranks reveal benchmark exposure?

Two experiments compare an OpenWebText-trained GPT-2 with MiniCPM5 Base. Both use teacher-forced reconstruction of complete space-delimited words at top-1, top-5 and top-10: first against familiar and novel controls, then across OLMES benchmark dates.

Research question

If a model encountered benchmark questions during training, it might predict their exact wording unusually well. We test whether OLMES receives an advantage beyond what general language ability produces, and whether the newer model loses that advantage on later benchmark families.

Predict the question, not the answer

Answer options and correct answers are excluded. Each model receives the first half of a text and predicts the words in the remaining half. Scoring is teacher forced: before every prediction, the model sees all preceding true words. For BoolQ and SocialQA, the text contains context followed by the question, so some predicted words are context.

Outcomes: exact full-word top-1, top-5 and top-10 reconstruction. A space-delimited word, including attached punctuation, is a hit only when every tokenizer token composing it is within the corresponding next-token cutoff. This is not question-answering accuracy.

Why use word ranks?

Raw negative log probability is sensitive to calibration. Two models can order candidates identically while assigning different probability values because one has sharper logits. Top-k accuracy depends not only on order, so additive logit shifts and positive temperature rescaling cannot change it.

At every token inside a target word, teacher forcing supplies the complete true prefix. Top-1 reconstructs the word only when every constituent token is the model's first choice. At top-5 or top-10, the true token path must survive that cutoff at every step until the complete word has been reconstructed.

This includes **every continuation word**, including multi-token words. It is an exact path-survival test, not a claim that the sequence is globally one of the k highest-probability complete words.

Why report three cutoffs? Agreement across top-1, top-5 and top-10 would be stronger evidence than selecting whichever k looks favorable. Rank at each step is invariant to additive logit shifts and positive temperature rescaling.

Tradeoff: a tokenizer that splits a word into more pieces creates more chances to fail. On OLMES, GPT-2 uses 1.27 tokens per word and MiniCPM 1.21. Rank removes probability calibration, but not tokenizer fragmentation, intelligence, model size, domain knowledge or training volume.

Models

Older: Stanford GPT-2 Small, 124M parameters, documented as trained on OpenWebText. Newer: MiniCPM5-1B-Base, 1.08B parameters, whose mixture includes Ultra-FineWeb and other datasets. Both pinned base checkpoints predate the IOL 2026 control. They are not a controlled architecture pair.

Experiment 1 / Design and controls

Question: does OLMES wording receive an unusually large word-prediction advantage relative to familiar corpus excerpts and genuinely recent human-written text?

OLMES: 30 items, three from each of ARC Easy, ARC Challenge, BoolQ, CommonsenseQA, HellaSwag, OpenBookQA, PIQA, SocialIQA, WinoGrande and MMLU.

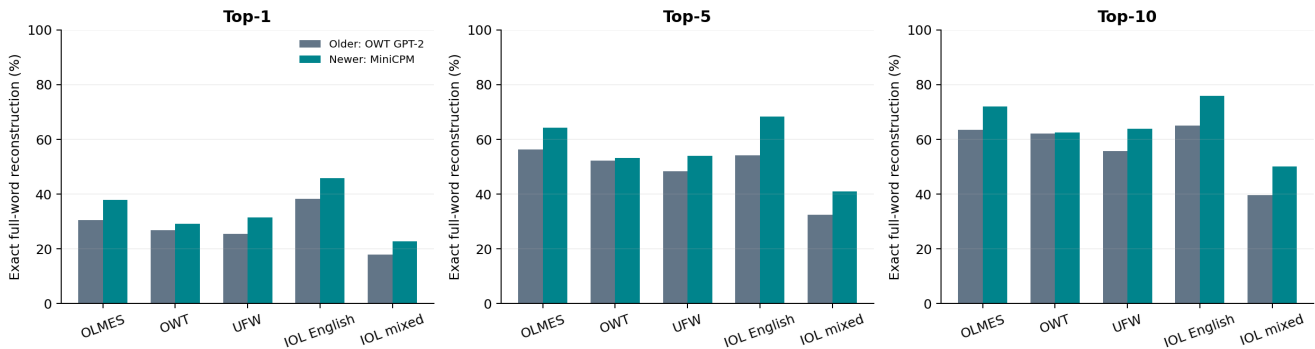
Corpus controls: 30 OpenWebText and 30 Ultra-FineWeb excerpts, roughly matched to OLMES length. They are verified corpus members, but exact consumption by the selected checkpoints is unverified and they are not genre matched to exam questions.

Post-checkpoint control: 30 non-overlapping mixed-text excerpts from five human-composed IOL 2026 problems. Because unfamiliar languages and tables make those excerpts a poor format match, ten short English-prose excerpts from the same five problems form the main sensitivity control.

The rank experiment covers 130 natural-language texts. Fresh SHA-256 strings from an earlier diagnostic are omitted because each is one space-delimited segment with no next-word position. IOL has five independent parent-problem clusters, not 30 independent questions.

Experiment 1 / Control results

Teacher-forced reconstruction of complete space-delimited words



OLMES: GPT-2 reconstructs 30.6%, 56.4% and 63.4% at top-1, top-5 and top-10. MiniCPM reaches 37.9%, 64.2% and 72.1%. Its gains are +7.3, +7.9 and +8.7 percentage points; paired cluster-bootstrap intervals exclude zero in this pilot.

Relative to each model's own corpus control, the OLMES advantage rises by 2.7 top-1 points, 6.0 top-5 points and 7.0 top-10 points. This is consistent with extra familiarity, but the corpus controls are not genre matched.

Crucial sensitivity: on novel English IOL, MiniCPM gains +7.5, +14.2 and +10.8 points—larger than its OLMES gains at all three cutoffs. General predictive ability can therefore produce at least as much improvement as the proposed familiarity signal.

Mixed IOL remains difficult because it contains unfamiliar languages and tables. English IOL is the more informative control. Experiment 1 does not isolate direct exposure.

Experiment 2 / Hypothesis and design

We next stay within OLMES and ask whether MiniCPM’s relative word-prediction advantage shrinks as benchmark date increases.

Specific prediction: if the newer model benefits disproportionately from exposure to older benchmark wording and that benefit fades for later families, newer-minus-older accuracy should have a negative date slope.

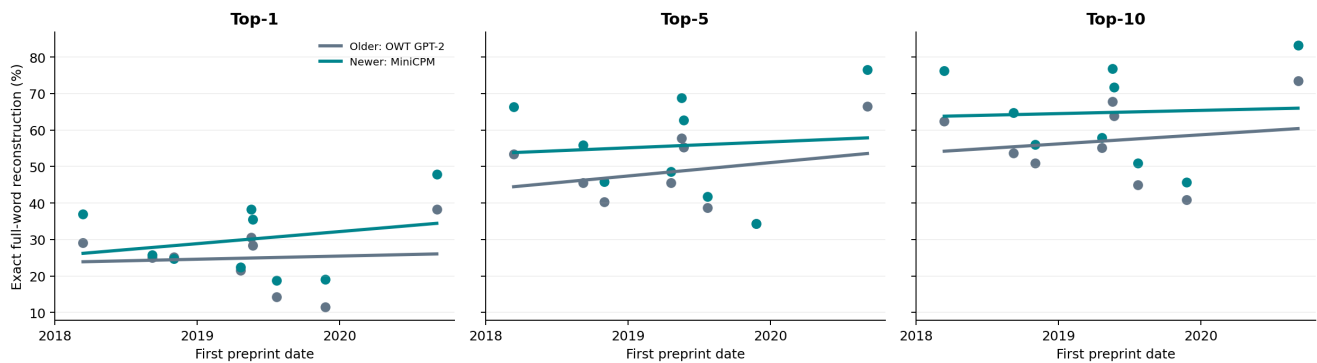
We froze a separate sample of 300 items, 30 per component. There are 296 paired texts with at least one continuation word and **4,699 complete word positions per model**. No multi-token word is discarded.

Dates are first arXiv submission dates from 2018–2020. They are benchmark-age proxies, not verified creation dates for individual questions. ARC Easy and Challenge share one paper and are averaged into one family, leaving nine equally weighted families. MMLU is the only 2020 family.

Expected if the hypothesis is true: MiniCPM’s advantage falls as date increases. A downward fitted line in the direct advantage plot supports that direction. Because every family predates MiniCPM, leakage need not produce this pattern.

Experiment 2 / Accuracy by benchmark date

Full-word reconstruction by benchmark date

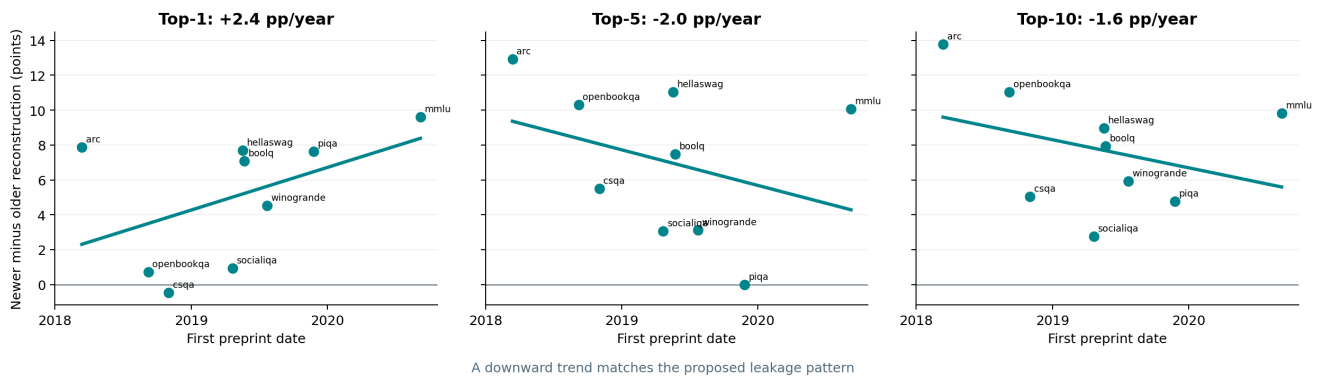


Both models' fitted reconstruction rises with date, but date is entangled with task composition. At top-1, GPT-2 rises +0.9 points/year and MiniCPM +3.3. At top-5 the slopes are +3.7 and +1.6; at top-10, +2.5 and +0.9.

These individual slopes do not isolate familiarity. The direct test subtracts GPT-2 accuracy from MiniCPM accuracy within each family, shown next.

Experiment 2 / Direct advantage slopes

Does the newer model lose its full-word advantage?



Observed: newer-minus-older reconstruction slopes +2.4 points/year at top-1, -2.0 at top-5 and -1.6 at top-10. Top-1 points opposite the prediction; top-5 and top-10 point with it.

The descriptive 95% family-bootstrap intervals are [-1.4, +7.5] for top-1, [-7.9, +2.4] for top-5 and [-6.2, +2.8] for top-10 percentage points/year. Every interval includes zero. The direction changes with k, and nine heterogeneous families provide limited precision.

Experiment 2 therefore does not detect the proposed temporal degradation. An interval crossing zero is not proof of equal slopes or absence of exposure.

Interpretation and reproducibility

Experiment 1: OLMES gets a meaningful newer-model reconstruction gain, but the novel English control gets a larger gain at every cutoff. **Experiment 2:** top-1 and top-5/top-10 disagree on temporal direction, and all intervals cross zero.

Per-token ranks solve the cross-model calibration objection. Full-word reconstruction includes every word. It does not solve tokenizer fragmentation or the intelligence confound. These results do not provide convincing evidence that leakage explains the model difference, and they do not establish that either model is contamination free.

Stronger follow-up: compare multiple years of the same exam series across a verified cutoff, match subject and format, include confirmed consumed records, and use models with a shared tokenizer.

Execution and saved evidence

FP16 inference ran locally on a GTX 1650 (4 GB). The final control and temporal passes took 68.3 and 99.4 seconds. Cost: \$0. Five project scoring tests passed. Invariants verified exact decoded word spans, identical word counts and top-1 <= top-5 <= top-10.

Saved outputs include pilot group summaries, cluster contrasts, paired model advantages, temporal component means, family-bootstrap statistics and complete inference and analysis scripts. The public bundle excludes sampled source text and model weights.

Pinned GPT-2: 2f4e2079c9ac92c2b5c6fecc19fae645bcef56fa. Pinned MiniCPM: f120ef901e3a11c2ed5fcfd242296cea3e80191b. OLMES commit: 5a51f502d463b8cdc4a2dcad7d7096c41ff1197e.

Sources: [Patel and Han](#); [Stanford checkpoint documentation](#); [MiniCPM checkpoint](#); [OLMES](#); [IOL 2026 problems](#).